

ES -116

Principles and Applications of Electrical Engineering

Semester II, 2025-26

Response of Inductor, Capacitor in Electrical Circuits

Recap: Inductor & Capacitor Behaviour

→ Inductor resists the change in current.

$$v_L(t) = L di/dt$$

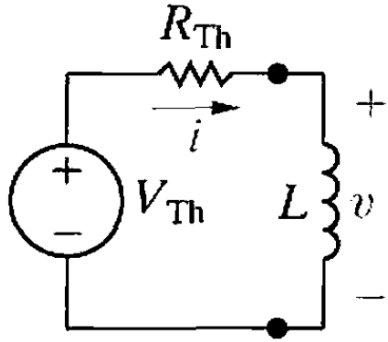
i.e. initially, it acts as an "open circuit" when it experiences a sudden change in the current. It becomes a "short-circuit" when a steady current starts to flow through it leading to zero voltage drop across it.

→ Capacitor resists the change in voltage.

$$i(t) = C dv_C/dt$$

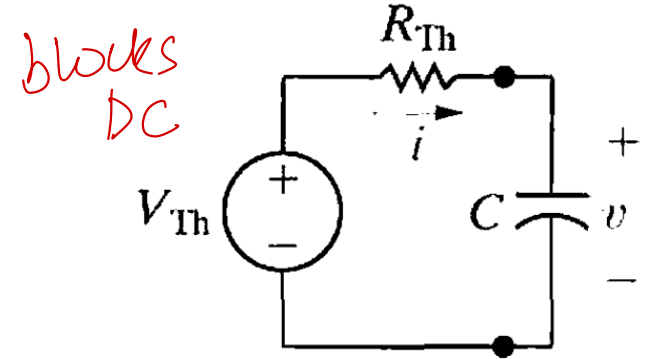
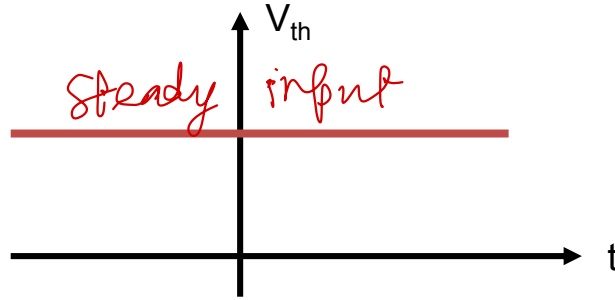
i.e. initially, it acts as a "short circuit" when it experiences a sudden change in voltage. It becomes an "open circuit" when a steady voltage is developed across it, leading to zero current.

Recap: Response to DC sources



$L \rightarrow$ short circuit

$$i = \frac{V_{Th}}{R_{Th}}$$

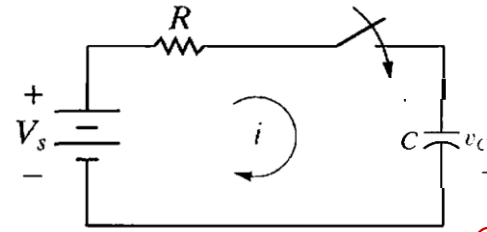
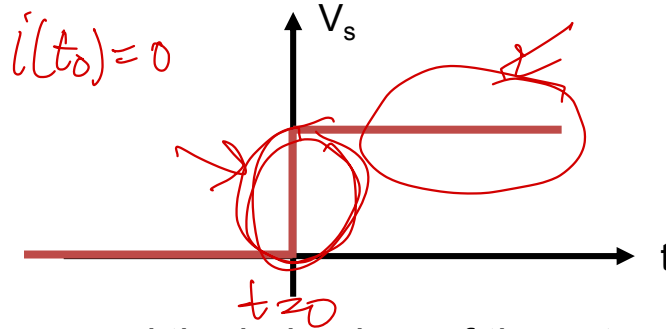
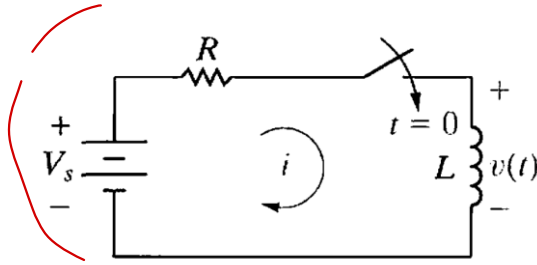


blocks
DC

open circuit

$$i = 0$$

Recap: Response to transient sources

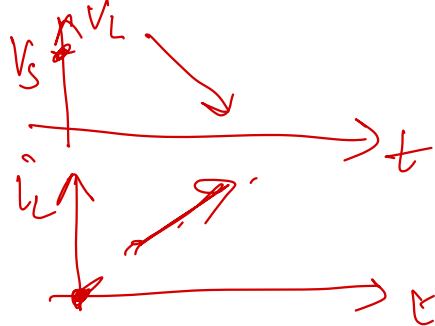


$$v_c(t_0) = 0$$

↓ s.c. at $t=0$

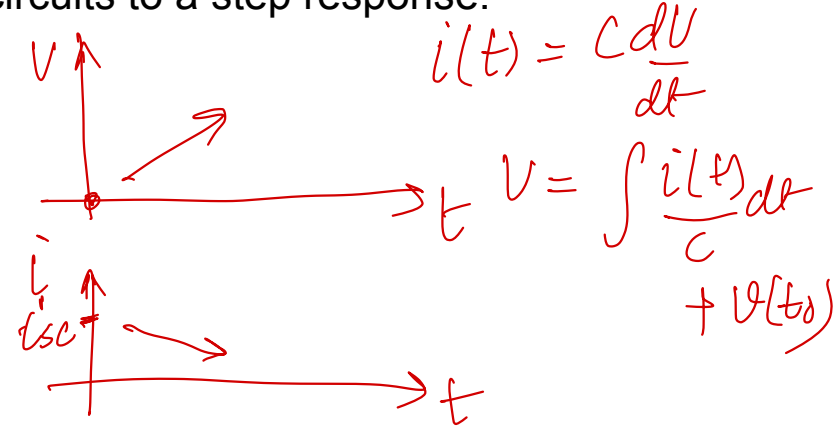
- In previous lecture, we discussed the behaviour of these two circuits to a step response.

at $t=0$, $L \rightarrow \infty$ & $C \rightarrow 0$



$$v = L \frac{di}{dt}$$

$$i(t) = \int_0^t \frac{v(t)}{L} dt + i(t_0)$$

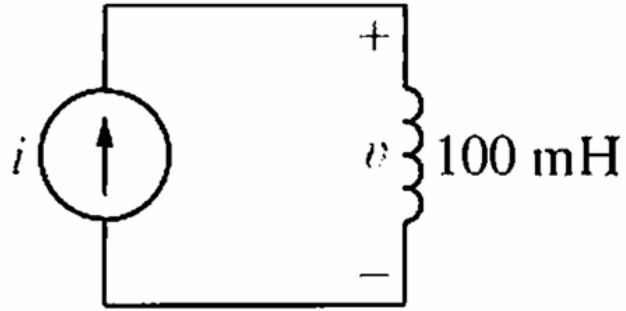


$$i(t) = C \frac{dv}{dt}$$

$$v = \int \frac{i(t)}{C} dt + v(t_0)$$

- In today's lecture, we will look at it mathematically, how would the response of these circuits look like at each point in time? → *Before that, let's calculate the response of individual response of these components.*

Response of Inductor to non-shocking Current Source



$$i = 0, \quad t < 0$$

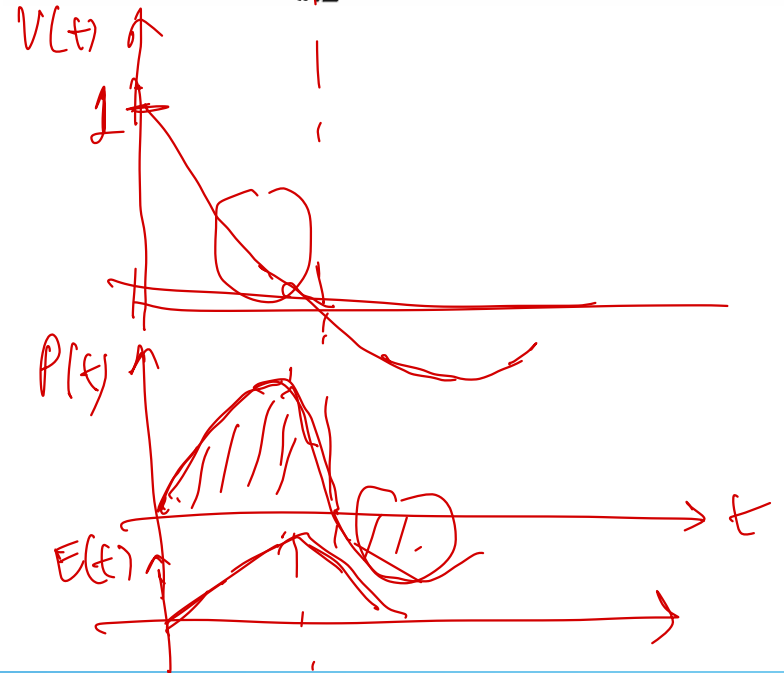
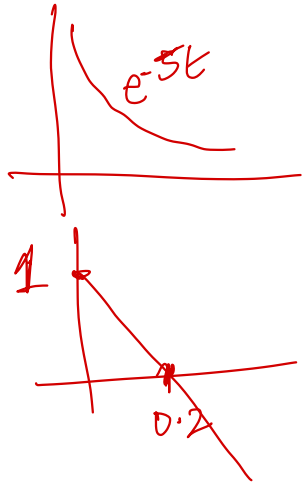
$$i = 10te^{-5t} \text{ A}, \quad t > 0$$

$$v(t) = L \frac{di}{dt}$$

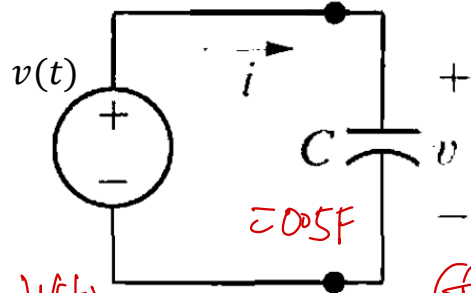
$$= e^{-5t}(1 - 5t)$$

$$P(t) = v(t) i(t)$$

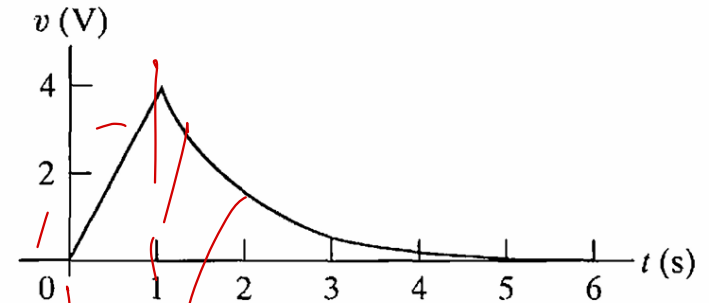
$\uparrow$ $\uparrow$
 $v(t)$ $i(t)$



Response of Capacitor to non-shocking Voltage Source



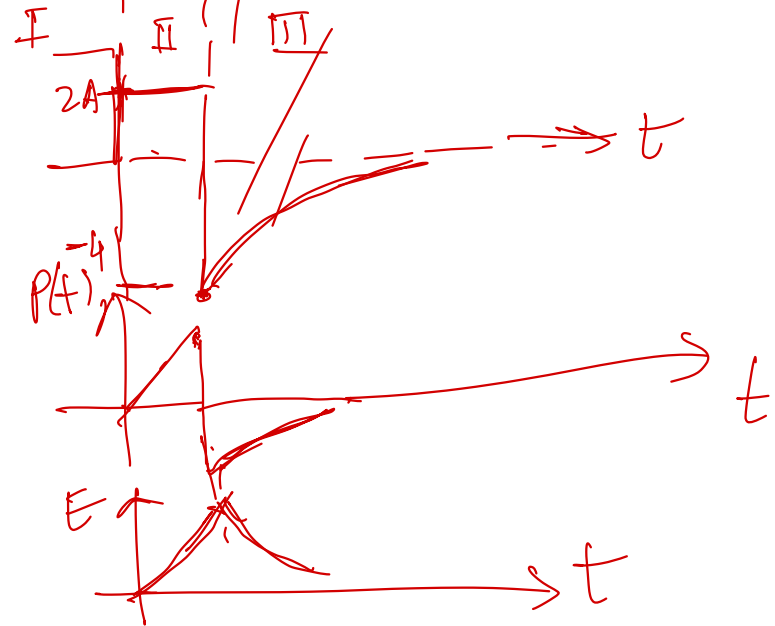
$$v(t) = \begin{cases} 0, & t \leq 0 \text{ s}; \\ 4t \text{ V}, & 0 \text{ s} \leq t \leq 1 \text{ s}; \\ 4e^{-(t-1)} \text{ V}, & t \geq 1 \text{ s}. \end{cases}$$



(I) $V(t) = 0$, $i(t) = C \frac{dV(t)}{dt} = 0$

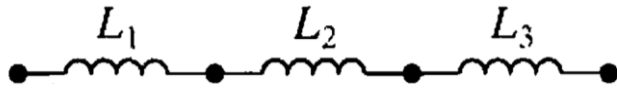
(II) $V(t) = 4t \text{ V}$, $i(t) = C \times 4 = 2 \text{ A}$

(III) $V(t) = 4e^{-(t-1)}$, $i(t) = -4e^{1-t}$
 $t \geq 1$, $i(t) = -\frac{4}{e^{t-1}}$

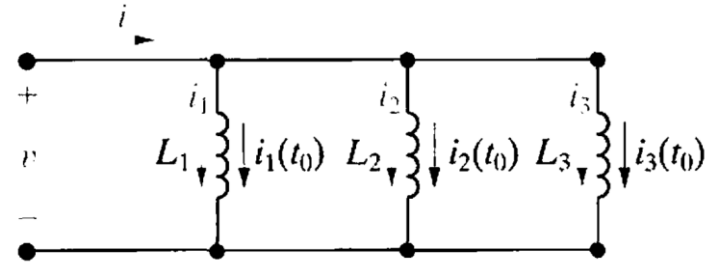


Eq. value if Inductors are put in Series/Parallel

How can we calculate the equivalent inductor value if multiple inductors are put in Series/Parallel?



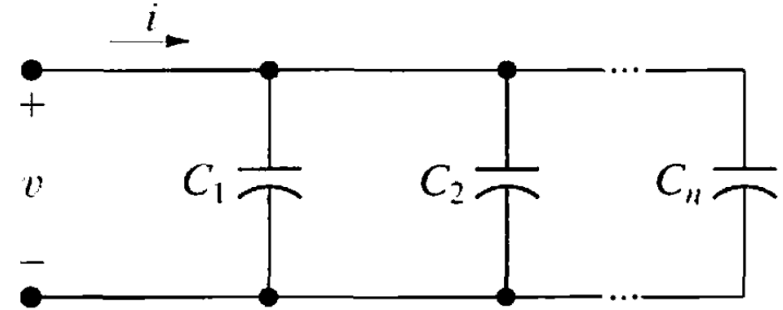
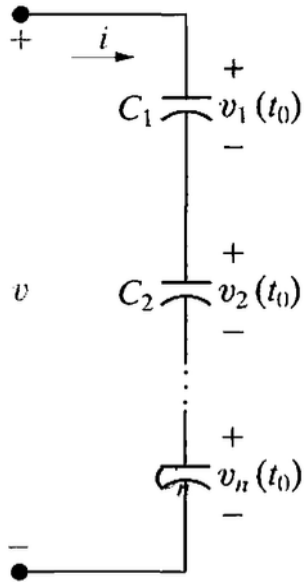
$i(t_0)$: Initial current



Eq. value if Capacitors are put in Series/Parallel

How can we calculate the equivalent capacitor value if multiple capacitors are put in Series/Parallel?

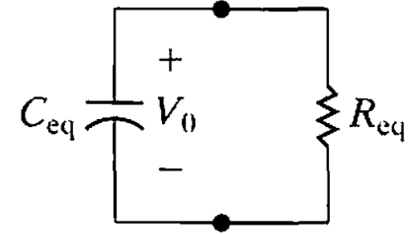
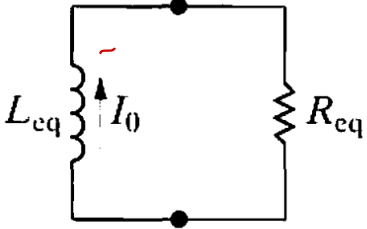
$v(t_0)$: Initial voltage



Natural Response of the RL, RC circuits

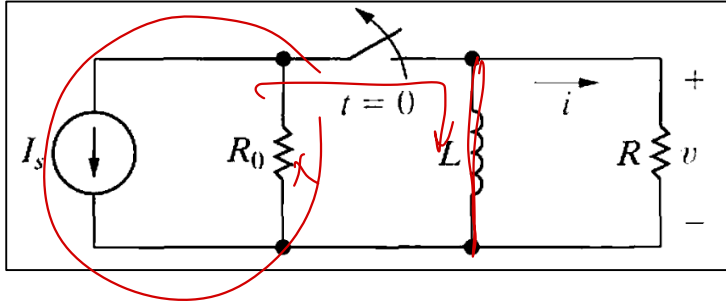
Let's first see the natural response of R-L and R-C circuit (with initial conditions).

How does the stand-alone system behave?



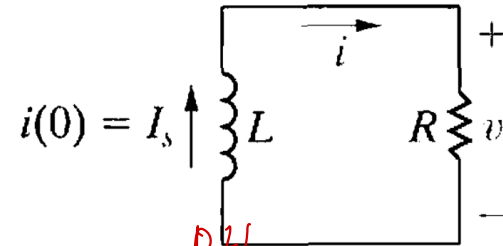
Natural Response of the RL circuit

The initial condition (stored energy) can be seen coming from the complete circuit.



When the switch is opened at $t=0$

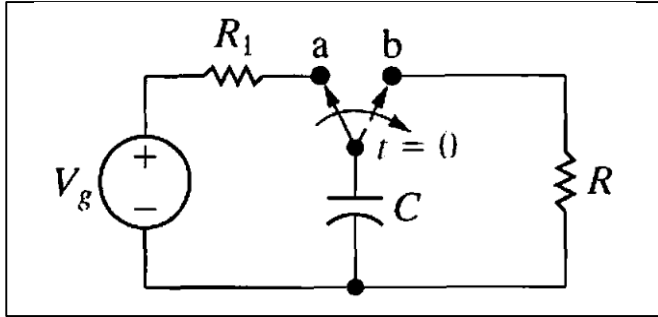
$$\tau = \frac{L}{R} \rightarrow \text{}$$



$$\begin{aligned} i(t) &= I_s \cdot e^{-Rt/L} \\ &= I_s e^{-t(R/L)} \\ &= I_s e^{-t/\tau} \end{aligned}$$

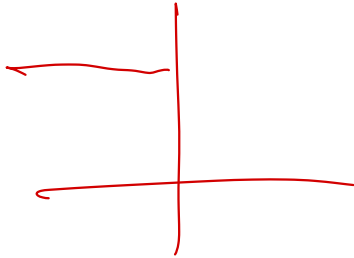
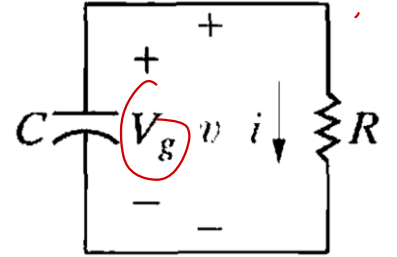
Natural Response of the RC circuit

The initial condition (stored energy) can be seen coming from the complete circuit.

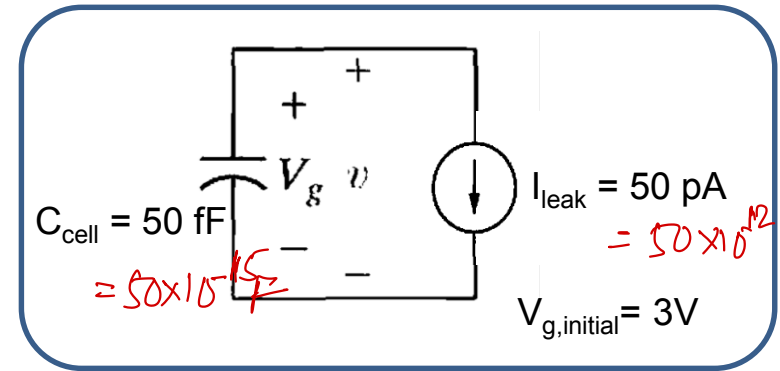
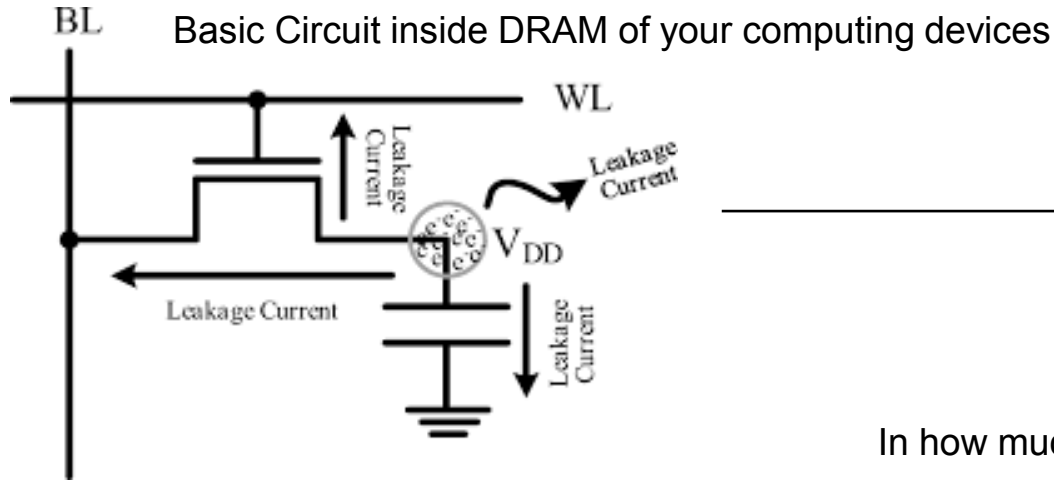


When the switch moves from a to b at $t=0$

$$\tau = RC$$



Practical Problem due to Leakage in Cap



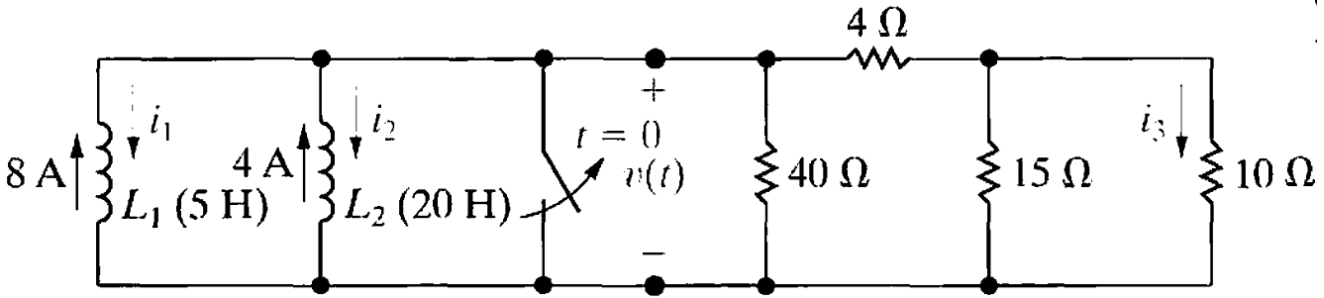
In how much time, the voltage across the capacitor will reduce to 1.5 V?

$$i(t) = C \frac{dV(t)}{dt}$$

$$\int_{V_{g,\text{ini}}}^{V(t)} V(t) = \int_0^t \frac{i(t)}{C} dt = \int_0^t \frac{I_{\text{leak}}}{C} dt$$

$$V(t) - V_{g,\text{ini}} = \frac{I_{\text{leak}} t}{C}$$

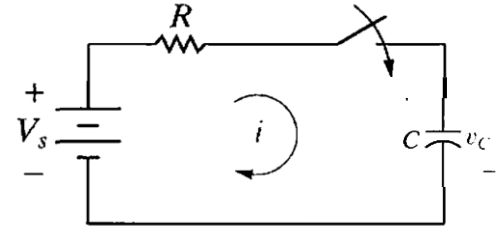
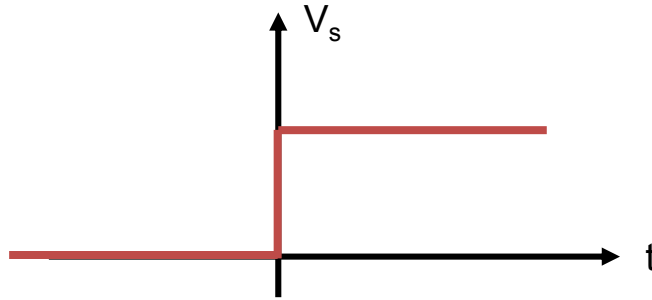
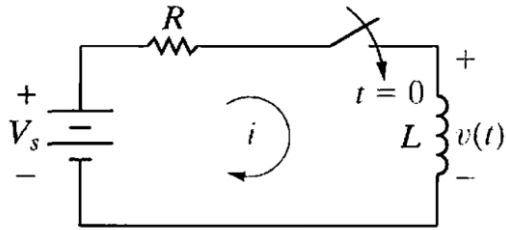
(Take Home) Question



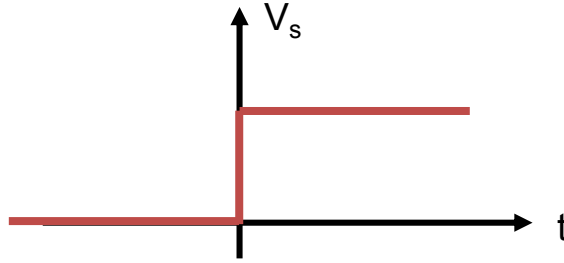
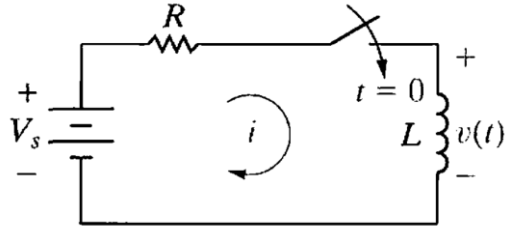
Hint: You can use series-parallel equivalent values for both resistors and inductors.

Step Response of the RL, RC circuits

Response with first order RL and RC circuit.



Step Response of the RL circuit



$$V_s(t) = i(t)R + L \frac{di(t)}{dt}$$

$$i(t) = \frac{V_s}{R} (1 - e^{-t/\tau})$$

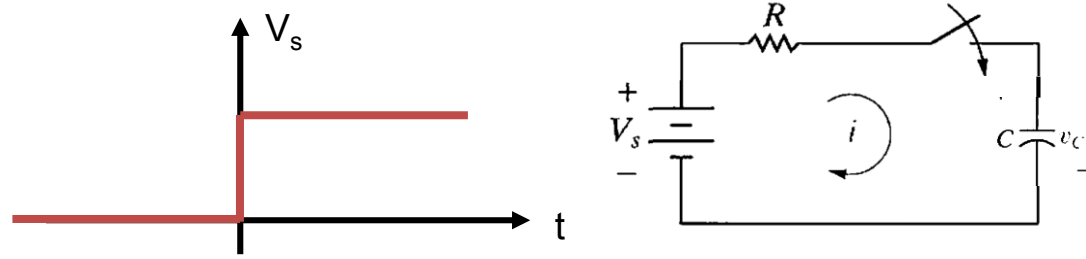
$$\text{at } t=0$$

$$t=\infty$$

$$\tau = \frac{L}{R}$$

Step Response of the RC circuit

if $V_C(0) = V_S$



$$V_s = iR + V_C$$

$$i = C \frac{dV_C}{dt}$$

$$V_s = RC \frac{dV_C(t)}{dt} + V_C(t)$$

$$V_C(t) = V_s (1 - e^{-t/\tau})$$

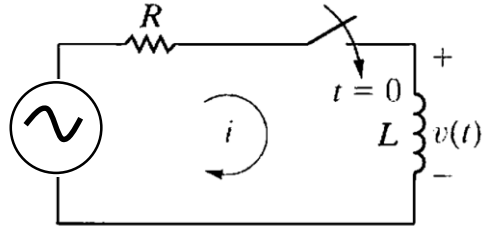
$$\tau = RC$$

at $t=0 \rightarrow V_C(t) = 0$

$t \rightarrow \infty, V_C(t) = V_s$

If time permits

Response to sinusoid of the RL circuit



Let's first understand physically what happens?

Response to Sinusoid of the RC circuit

Let's first understand physically what happens?

